The Second Law of Thermodynamics Ludwig Boltzmann, who spent much of his life studying s

Let's do some combinatorics here. Imagine you have a fair coin, and you flip that coin 3 times. You define an event E

How many possible outcomes are there inside the sample space Ω (or S)?

The event E is called as the **macrostate** of the system, and each of its elements is called **microstate**. For example, the ma
[width=0.7] /PhyAI/chapter2/2_1.*png* How many configurations satisfy 3 bonds containing 4 units of energy?   A solid conta
[width=0.7] /PhyAI/chapter2/2_2.*png* The general formula for the multiplicity of an Einstein solid This crucial formula is t

The problem is, N and q are always very large numbers. For example, in 1 gram of Na, the number of chemical bonds is up

$\ln N! = \ln\left[N(N-1)\cdots 2.1\right] = \ln N + \ln(N-1) + \cdots + \ln 2 + \ln 1$
$= \sum_{i=1}^{N} \ln i \approx \int_1^N \ln x dx \approx N\ln N - N$
[Stirling's Approximation] When $N >> 1$:

Recall the definition of Gamma function, for every non-negative integer N, we always have: $\Gamma(N+1) = \int_0^{+\infty} x^N e^{-x}$
$= \int_0^{+\infty} e^{N\ln x} e^{-x} dx = \int_0^{+\infty} e^{N\ln x - x} dx$
$= \int_0^{+\infty} e^{N\ln x - x} dx$ Now we want to expand the equation by Taylor expansion, by setting an auxiliary variable $x = y + N =$
$= N\ln N\left(\frac{y}{N}+1\right) - (y+N)$
$= N\ln N + N\left(\frac{y}{N} - \frac{1}{2}.\frac{y^2}{N^2}\right) - y - N$
$= N\ln N - \frac{1}{2}.\frac{y^2}{N} - N$
After changing the variable, we obtain: $\Gamma(N+1) = \int_0^{+\infty} e^{N\ln x - x} dx = \int_{-N}^{+\infty} \exp\left(N\ln N - \frac{1}{2}.\frac{y^2}{N} - N\right) dy$
$= N^N e^{-N} \int_{-N}^{+\infty} \exp\left(-\frac{1}{2}.\frac{y^2}{N}\right) dy$ Since N approaches infinity, applying Gaussian integral result:

We conclude: $\Gamma(N+1) = N^N e^{-N}\sqrt{2N}\int_{-\infty}^{+\infty} \exp\left[-\left(\frac{y}{\sqrt{2N}}\right)^2\right] d\left(\frac{y}{\sqrt{2N}}\right)$
$= N^N e^{-N}\sqrt{2\pi N}$ Let's apply our previous results to explore a very cool phenomenon. I have an **interacting system**, wh
[width=0.70] /PhyAI/chapter2/2_3.*png* What is the most probable configuration for energy distributing system?   We have t

We already assumed that $N_A = 6$, $N_B = 9$, and $q = q_A + q_B = 8$, so:

The multiplicity of an interacting system is: $\Omega_{AB} = \Omega_A.\Omega_B$

As you can see here, the most probable configuration of an interacting system is $q_A = 3$, $q_B = 5$, with the highest multiplici

Our calculation above seems like make sense; the "equilibrium" state of energy is the most likely reached state. Energy is w

The amount of energy stored is directly proportional to the number of chemical bonds in a solid; to put it more simply, the

Take natural logarithm of both sides, and use Theorem : $\ln\Omega(N,q) = \ln(q+N)! - \ln q! - \ln N! = (q+N)\ln(q+N) - ($
$= (q+N)\ln(q+N) - q\ln q - N\ln N$ Taking natural logarithm of the multiplicity of an interacting system yields: $\ln\Omega$